

Bryan-Elliott Tam

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Software engineer with an R&D background in symbolic AI, knowledge representation, and query engines. Builds across the stack, including web, distributed data, and embedded systems. Open-source work and side projects at imaginative-twilight-1fbea5.netlify.app/project.

SKILLS

- **Programming Languages:** TypeScript, Rust, Python, C++, Go, Prolog, SMT-LIB, Bash, SPARQL
- **Technologies:** Git, RDF, Svelte, Vue.js, React, Docker, MongoDB, Neo4j
- **Natural Languages:** French (Native), English (Fluent), Dutch (Basic)

WORK EXPERIENCES

Software Engineer (Decentralized Databases) September 2022 — Present
Ghent University – Imec *Gent, Belgium*

- Built scalable query-processing features for [Comunica](#), a widely-used open-source SPARQL query engine (TypeScript), for querying large decentralized datasets; published in peer-reviewed venues.
- Mentored master's students and interns; supported the Knowledge Graphs course.
- Program committee member for the [DMKG 2026 workshop](#). Contributed to the [SEMANTiCS 2025 Developers Workshop](#) committee and the [TREE W3C Community Group](#).

Co-founder & Software Engineer October 2025 — October 2026
VATmiraal *Gent, Belgium*

- Built an AI product for VAT determination and auditing end to end; incubated by [KBC Start it!](#).
- Designed a neurosymbolic engine combining a Prolog rule system with LLMs, served over an API and MCP server.
- Handled sales and business development.

Software Engineer – Research Internship September 2025 — October 2025
Inria *Nice, France*

Developed query-optimization techniques for federated queries in [Comunica](#) using RDF data shapes (doctoral internship, FWO-funded).

Software Engineer – Research Contracts Summer 2020 & 2022
Université Laval *Sainte-Foy, Qc, Canada*

- Built a custom literature search engine for architecture researchers: Go backend, Python recommender system, JavaScript frontend (2022).
- Extended a 2D excavator localization system to 3D, improving positional accuracy using map data and turret orientation; C++ and Python (2020).

IoT and Web Developer August 2018 — March 2020
Systèmes Vireo *Sainte-Foy, Qc, Canada*

- Led development and deployment of an end-to-end IoT platform for urban agriculture at an early-stage startup, from embedded firmware (C++, MQTT, KiCad) to full-stack web (React, TypeScript, MongoDB).
- Ran on-site hardware and software installation and testing.

EDUCATION

Ghent University 2022 — Present
Doctorate, Computer Science Engineering *Gent, Belgium*

Université Laval 2020 — 2022
Master of Sciences, Computer Science *Sainte-Foy, Qc, Canada*

Université Laval 2015 — 2019
Bachelor of Engineering, Mechanical Engineering *Sainte-Foy, Qc, Canada*